

Initialisation and reproducibility

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Regularisation and early stopping

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning rate experiments

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Updating parameters

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Distinguish equality of loss and parameters for Initialisation and reproducibility. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Initialisation and reproducibility. Proposed explanation: “The same seed specifies the entire experiment”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Transfer the starting-value experiment independently

Use $J(a) = (a - 3)^2$, gradient $2(a - 3)$, rate 0.25 and two updates. Restart from $a = 1$, then restart from $a = 5$. Calculate both traces and final losses. Explain what can be concluded about these two runs.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace a fresh starting value

For the scalar lesson model, start $v = -2$ and apply two updates at rate 0.1. Give all three v values and the final training MSE.

My reasoning and evidence

6. W2 Distinguish equality of loss and parameters

The lesson runs starting at 1 and 3 finish with the same training MSE. Give their final v values and explain the equality without calling the models identical.

My reasoning and evidence

7. W3 Audit a seed claim

Both runs use the same seed and generator. Run A assigns the first four draws to its weights. Run B draws a preview number first, then assigns the next four. Must the initial weight arrays match? What should be saved?

My reasoning and evidence

8. W4 Transfer the starting-value experiment independently

Use $J(a) = (a - 3)^2$, gradient $2(a - 3)$, rate 0.25 and two updates. Restart from $a = 1$, then restart from $a = 5$. Calculate both traces and final losses. Explain what can be concluded about these two runs.

My reasoning and evidence

9. W5 Test repeated correctness

A forward-pass program returns the same wrong answer every run with a fixed seed. Has repeatability verified the implementation? Name two stronger checks.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled initialisation and reproducibility evidence sheet to Image-pattern Network Investigation. Show how you can trace how starting values affect a fixed training budget. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Initialisation and reproducibility

Reproduce two training traces by saving their distinct initial values.

$\text{LossJ}(a) = (a-2)^2$; gradient $2(a-2)$; rate 0.25; two updates. Restart from $a = 0$ and separately $a = 4$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Initialisation and reproducibility

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Start from 0

Updates: $a = 1$ then 1.5; final loss 0.25.



2. Start from 4

Updates: $a = 3$ then 2.5; final loss 0.25.



3. Record experiment

Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

CHANGE ONE THING

Repeat the $a = 0$ run with identical arithmetic and settings.

Explain the changed result using the matching lesson.

Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Reproduce two training traces by saving their distinct initial values.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

1. Record experiment
2. Start from 0
3. Start from 4

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Record experiment Save initial values, rate, objective and update count; the same final loss hides different parameter paths. Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts. Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

Practice 2: explain the deciding evidence

Changed-case practice: Repeat the $a = 0$ run with identical arithmetic and settings. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.

3. It again reaches 1 then 1.5 in this deterministic toy calculation.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: It again reaches 1 then 1.5 in this deterministic toy calculation. Identical initial state and update rules determine the same arithmetic trace.

Repair a missing building block

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from prediction error. Short practice: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
